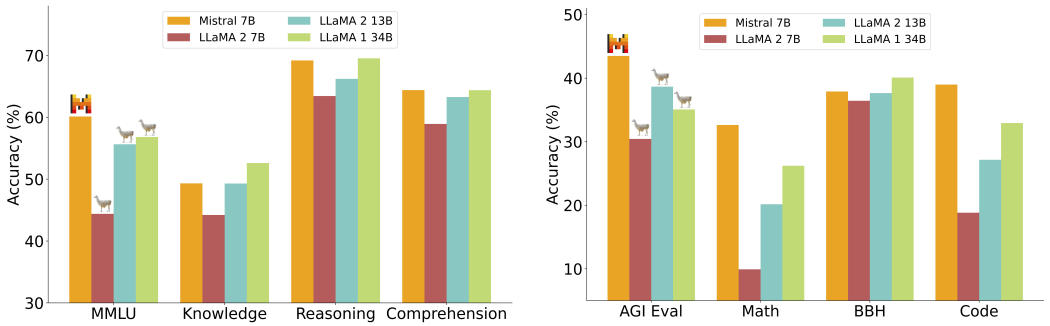


Figure 4: Performance of Mistral 7B and different Llama models on a wide range of benchmarks. All models were re-evaluated on all metrics with our evaluation pipeline for accurate comparison. Mistral 7B significantly outperforms Llama 2 7B and Llama 2 13B on all benchmarks. It is also vastly superior to Llama 1 34B in mathematics, code generation, and reasoning benchmarks.

Model	Modality	MMLU	HellaSwag	WinoG	PIQA	Arc-e	Arc-c	NQ	TriviaQA	HumanEval	MBPP	MATH	GSM8K
LLaMA 2 7B	Pretrained	44.4%	77.1%	69.5%	77.9%	68.7%	43.2%	24.7%	63.8%	11.6%	26.1%	3.9%	16.0%
LLaMA 2 13B	Pretrained	55.6%	**80.7%**	72.9%	80.8%	75.2%	48.8%	**29.0%**	**69.6%**	18.9%	35.4%	6.0%	34.3%
Code-Llama 7B	Finetuned	36.9%	62.9%	62.3%	72.8%	59.4%	34.5%	11.0%	34.9%	**31.1%**	**52.5%**	5.2%	20.8%
Mistral 7B	Pretrained	**60.1%**	**81.3%**	**75.3%**	**83.0%**	**80.0%**	**55.5%**	**28.8%**	**69.9%**	**30.5%**	47.5%	**13.1%**	**52.2%**

Table 2: Comparison of Mistral 7B with Llama. Mistral 7B outperforms Llama 2 13B on all metrics, and approaches the code performance of Code-Llama 7B without sacrificing performance on non-code benchmarks.

Size and Efficiency. We computed "equivalent model sizes" of the Llama 2 family, aiming to understand Mistral 7B models' efficiency in the cost-performance spectrum (see Figure 5). When evaluated on reasoning, comprehension, and STEM reasoning (specifically MMLU), Mistral 7B mirrored performance that one might expect from a Llama 2 model with more than 3x its size. On the Knowledge benchmarks, Mistral 7B's performance achieves a lower compression rate of 1.9x, which is likely due to its limited parameter count that restricts the amount of knowledge it can store.

Evaluation Differences. On some benchmarks, there are some differences between our evaluation protocol and the one reported in the Llama 2 paper: 1) on MBPP, we use the hand-verified subset 2) on TriviaQA, we do not provide Wikipedia contexts.

4 Instruction Finetuning

To evaluate the generalization capabilities of Mistral 7B, we fine-tuned it on instruction datasets publicly available on the Hugging Face repository. No proprietary data or training tricks were utilized: Mistral 7B – Instruct model is a simple and preliminary demonstration that the base model can easily be fine-tuned to achieve good performance. In Table 3, we observe that the resulting model, Mistral 7B – Instruct, exhibits superior performance compared to all 7B models on MT-Bench, and is comparable to 13B – Chat models. An independent human evaluation was conducted on https://llmboxing.com/leaderboard.

Model	Chatbot Arena ELO Rating	MT Bench
WizardLM 13B v1.2	1047	7.2
Mistral 7B Instruct	**1031**	**6.84 +/- 0.07**
Llama 2 13B Chat	1012	6.65
Vicuna 13B	1041	6.57
Llama 2 7B Chat	985	6.27
Vicuna 7B	997	6.17
Alpaca 13B	914	4.53

Table 3: Comparison of Chat models. Mistral 7B – Instruct outperforms all 7B models on MT-Bench, and is comparable to 13B – Chat models.

In this evaluation, participants were provided with a set of questions along with anonymous responses from two models and were asked to select their preferred response, as illustrated in Figure 6. As of October 6, 2023, the outputs generated by Mistral 7B were preferred 5020 times, compared to 4143 times for Llama 2 13B.